

Assignment 1\A1_Code_2.py

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from matplotlib.animation import FuncAnimation
4 import time
5
6 start_time1 = time.perf_counter()
7
8 x_start = 0
9 x_end = 1
10
11 t_start = 0
12 t_end = 1
13
14 nx = 50 #number of grid points for x
15 CFL = 0.8
16
17 a = 1
18
19 dx = (x_end-x_start)/nx
20 dt = (CFL*dx)/a
21
22 nt = int((t_end-t_start)/dt)
23
24 x50 = np.arange(x_start,x_end,dx)
25
26 u0 = np.exp(-100*(x50-0.5)**2)
27
28 u_n = np.zeros(len(x50))
29 u_np1 = np.zeros(len(x50))
30 u_history = np.zeros((nt,nx))
31
32
33 u_n = u0.copy()
34
35 for n in range(nt):
36     for i in range(nx):
37         if i == 0:
38             u_np1[i] = u_n[i] - CFL*(u_n[i] - u_n[-1])
39         else:
40             u_np1[i] = u_n[i] - CFL*(u_n[i] - u_n[i-1])
41     u_n = u_np1.copy()
42     u_history[n,:] = u_n.copy()
43
44 u_history50 = u_history
45
46 end_time1 = time.perf_counter()
47
48 time1 = start_time1-end_time1
```

```
49
50 start_time2 = time.perf_counter()
51
52 x_start = 0
53 x_end = 1
54
55 t_start = 0
56 t_end = 1
57
58 dx50 = dx
59 e50 = 1/nx * np.sum(np.abs(u_history50[-1,:] - u_history50[0,:]))
60
61 nx = 100 #number of grid points for x
62 CFL = 0.8
63
64 a = 1
65
66 dx = (x_end-x_start)/nx
67 dt = (CFL*dx)/a
68
69 nt = int((t_end-t_start)/dt)
70
71 x100 = np.arange(x_start,x_end,dx)
72
73 u0 = np.exp(-100*(x100-0.5)**2)
74
75 u_n = np.zeros(len(x100))
76 u_np1 = np.zeros(len(x100))
77 u_history = np.zeros((nt,nx))
78
79
80 u_n = u0.copy()
81
82 for n in range(nt):
83     for i in range(nx):
84         if i == 0:
85             u_np1[i] = u_n[i] - CFL*(u_n[i] - u_n[-1])
86         else:
87             u_np1[i] = u_n[i] - CFL*(u_n[i] - u_n[i-1])
88     u_n = u_np1.copy()
89     u_history[n,:] = u_n.copy()
90
91 u_history100 = u_history
92
93 end_time2 = time.perf_counter()
94
95 time2 = start_time2-end_time2
96
97 start_time3 = time.perf_counter()
98
```

```
99  dx100 = dx
100
101  e100 = 1/nx * np.sum(np.abs(u_history100[-1,:] - u_history100[0,:]))
102
103
104  x_start = 0
105  x_end = 1
106
107  t_start = 0
108  t_end = 1
109
110  nx = 200 #number of grid points for x
111  CFL = 0.8
112
113  a = 1
114
115  dx = (x_end-x_start)/nx
116  dt = (CFL*dx)/a
117
118  nt = int((t_end-t_start)/dt)
119
120  x200 = np.arange(x_start,x_end,dx)
121
122  u0 = np.exp(-100*(x200-0.5)**2)
123
124  u_n = np.zeros(len(x200))
125  u_np1 = np.zeros(len(x200))
126  u_history = np.zeros((nt,nx))
127
128
129  u_n = u0.copy()
130
131  for n in range(nt):
132      for i in range(nx):
133          if i == 0:
134              u_np1[i] = u_n[i] - CFL*(u_n[i] - u_n[-1])
135          else:
136              u_np1[i] = u_n[i] - CFL*(u_n[i] - u_n[i-1])
137      u_n = u_np1.copy()
138      u_history[n,:] = u_n.copy()
139
140  u_history200 = u_history
141
142  dx200 = dx
143  e200 = 1/nx * np.sum(np.abs(u_history200[-1,:] - u_history200[0,:]))
144
145  end_time3 = time.perf_counter()
146
147  time3 = start_time3-end_time3
148
```

```
149 start_time4 = time.perf_counter()
150
151 x_start = 0
152 x_end = 1
153
154 t_start = 0
155 t_end = 1
156
157 nx = 400 #number of grid points for x
158 CFL = 0.8
159
160 a = 1
161
162 dx = (x_end-x_start)/nx
163 dt = (CFL*dx)/a
164
165 nt = int((t_end-t_start)/dt)
166
167 x400 = np.arange(x_start,x_end,dx)
168
169 u0 = np.exp(-100*(x400-0.5)**2)
170
171 u_n = np.zeros(len(x400))
172 u_np1 = np.zeros(len(x400))
173 u_history = np.zeros((nt,nx))
174
175
176 u_n = u0.copy()
177
178 for n in range(nt):
179     for i in range(nx):
180         if i == 0:
181             u_np1[i] = u_n[i] - CFL*(u_n[i] - u_n[-1])
182         else:
183             u_np1[i] = u_n[i] - CFL*(u_n[i] - u_n[i-1])
184     u_n = u_np1.copy()
185     u_history[n,:] = u_n.copy()
186
187 u_history400 = u_history
188
189 dx400 = dx
190 e400 = 1/nx * np.sum(np.abs(u_history400[-1,:] - u_history400[0,:]))
191
192 end_time4 = time.perf_counter()
193
194 time4 = start_time4-end_time4
195
196
197
198
```

```
199 plt.subplot(2,2,1)
200 plt.plot(x50,u_history50[0,:],"k",label="Initial Solution")
201 plt.plot(x50,u_history50[-1,:],"r",label="Final Solution")
202 plt.xlabel("x")
203 plt.ylabel("u")
204 plt.legend()
205 plt.title("n = 50")
206
207 plt.subplot(2,2,2)
208 plt.plot(x100,u_history100[0,:],"k",label="Initial Solution")
209 plt.plot(x100,u_history100[-1,:],"r",label="Final Solution")
210 plt.xlabel("x")
211 plt.ylabel("u")
212 plt.legend()
213 plt.title("n = 100")
214
215 plt.subplot(2,2,3)
216 plt.plot(x200,u_history200[0,:],"k",label="Initial Solution")
217 plt.plot(x200,u_history200[-1,:],"r",label="Final Solution")
218 plt.xlabel("x")
219 plt.ylabel("u")
220 plt.legend()
221 plt.title("n = 200")
222
223 plt.subplot(2,2,4)
224 plt.plot(x400,u_history400[0,:],"k",label="Initial Solution")
225 plt.plot(x400,u_history400[-1,:],"r",label="Final Solution")
226 plt.xlabel("x")
227 plt.ylabel("u")
228 plt.legend()
229 plt.title("n = 400")
230
231 plt.show()
232
233 n = [50, 100, 200, 400]
234 t = [-time1, -time2, -time3, -time4]
235 plt.plot(n,t)
236 plt.xlabel("Number of Nodes")
237 plt.ylabel("Computing Time")
238 plt.show()
239
240 e = [e50, e100, e200, e400]
241 dxvec = [dx50, dx100, dx200,dx400]
242 plt.plot(dxvec,e)
243 plt.xlabel("dx")
244 plt.ylabel("error")
245 plt.show()
```